Forecasting degree heating weeks with fourier neural operators for coral bleaching early warning

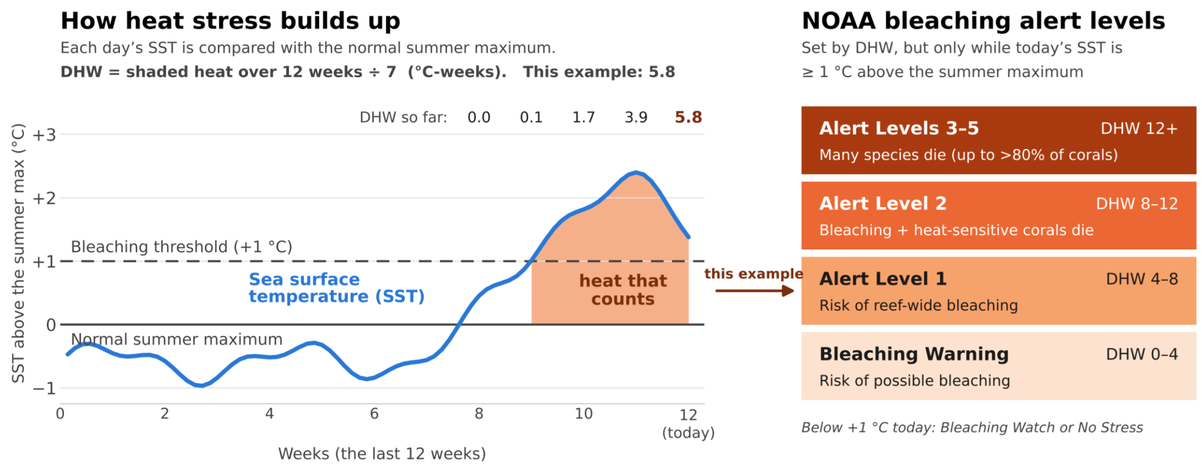
Xuan Ting Lei
College of Systems and Society, School of Computing, Australian National University

Summary

- A deep learning model forecasts global ocean temperature up to 90 days ahead.
- From those forecasts we predict coral heat stress, measured in Degree Heating Weeks (DHW)[1].
- It beats a persistence forecast at every lead time out to 90 days, this is state of the art.
- Takeaway: choose forecast models for heat-stress skill, not just temperature accuracy.

Introduction

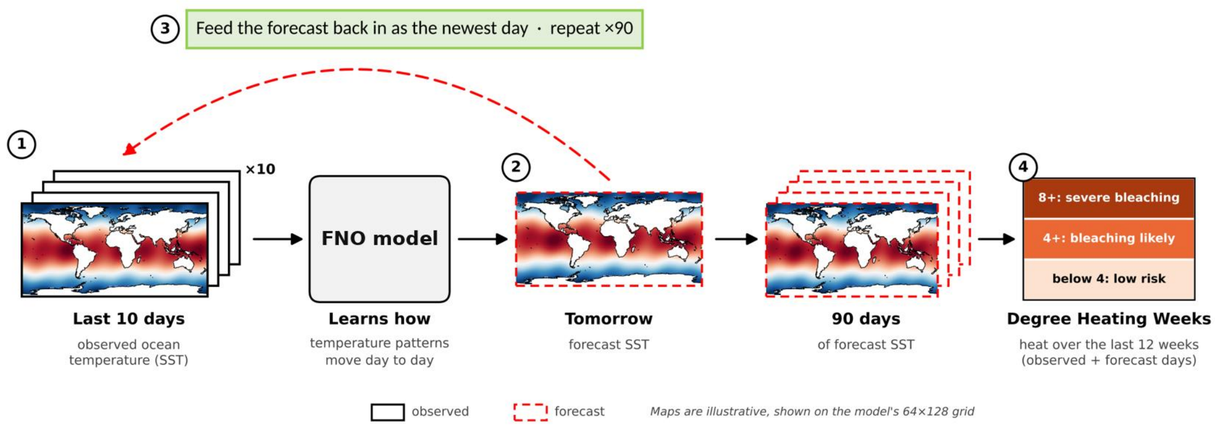
- Coral bleaching is a leading cause of global reef decline.
- Bleaching is triggered by prolonged ocean heat stress[2].
- DHW: heat built up above a reef's normal summer maximum over the past 12 weeks.



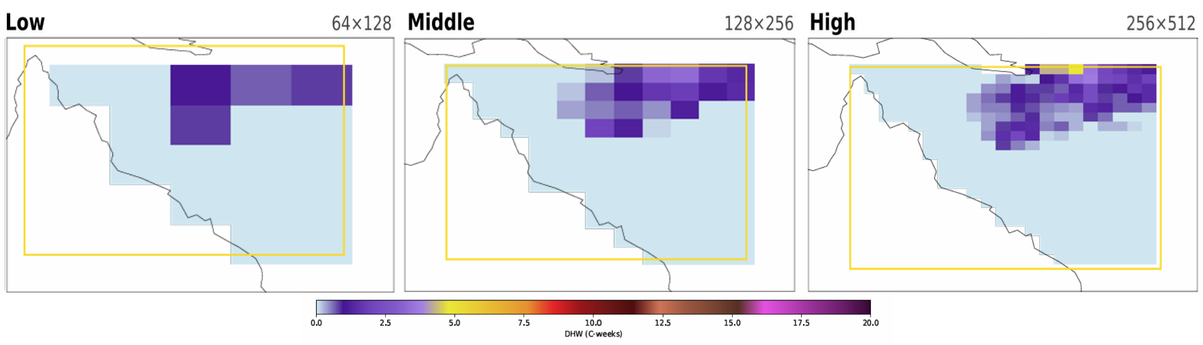
- Current outlooks rely on physics-based climate models that are costly to run.
- Deep learning forecasting of DHW is largely unexplored.
- Corals can recover from a single bleaching event if heat stress eases[3], so earlier warnings give managers time to act and test interventions.

Methods

- One of the first uses of a Fourier Neural Operator (FNO) to forecast DHW.
- What the FNO is: a deep learning model that learns how ocean temperature patterns change from day to day. It learns broad, large-scale patterns rather than single points, which suits smoothly varying ocean temperatures[4].
- It is fast to run once trained, making it suitable for operational weather forecasting.
- Two stages:
  - Forecast global sea surface temperature (SST) up to 90 days ahead.
  - Calculate DHW from the forecast temperatures.

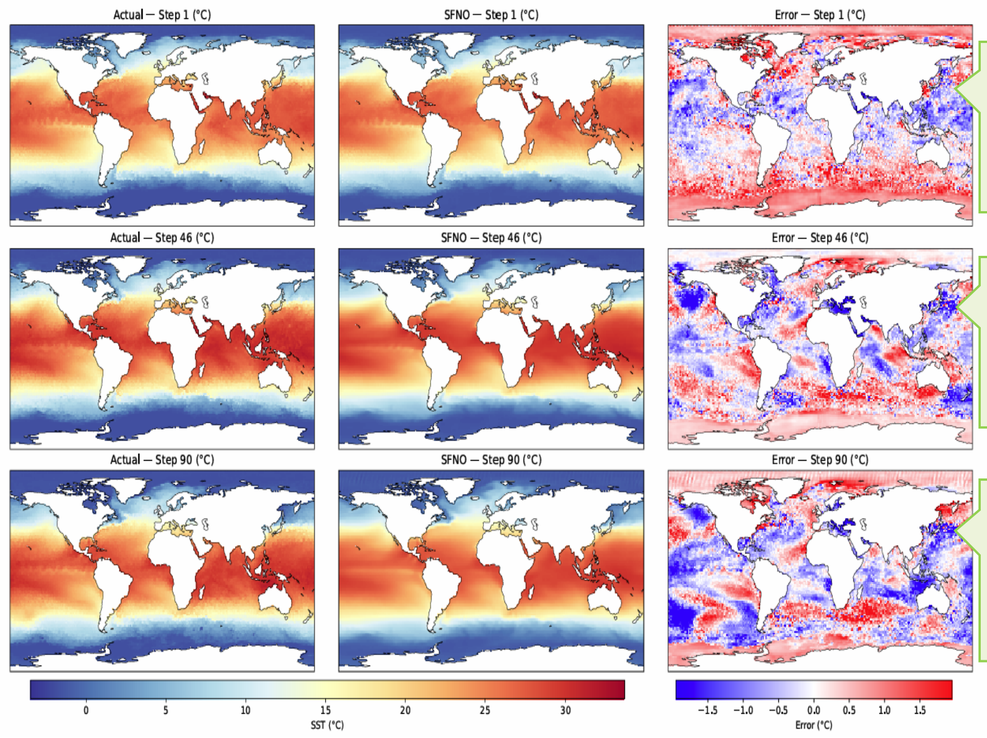


- Fourteen model configurations compared to find the best forecaster.
- Three map resolutions tested: 64x128 (Low), 128x256 (Middle), 256x512 (High).



DHW predictions at day 46 for the Great Barrier Reef at three resolutions - °C weeks

Sea Surface Temperature (SST) Results



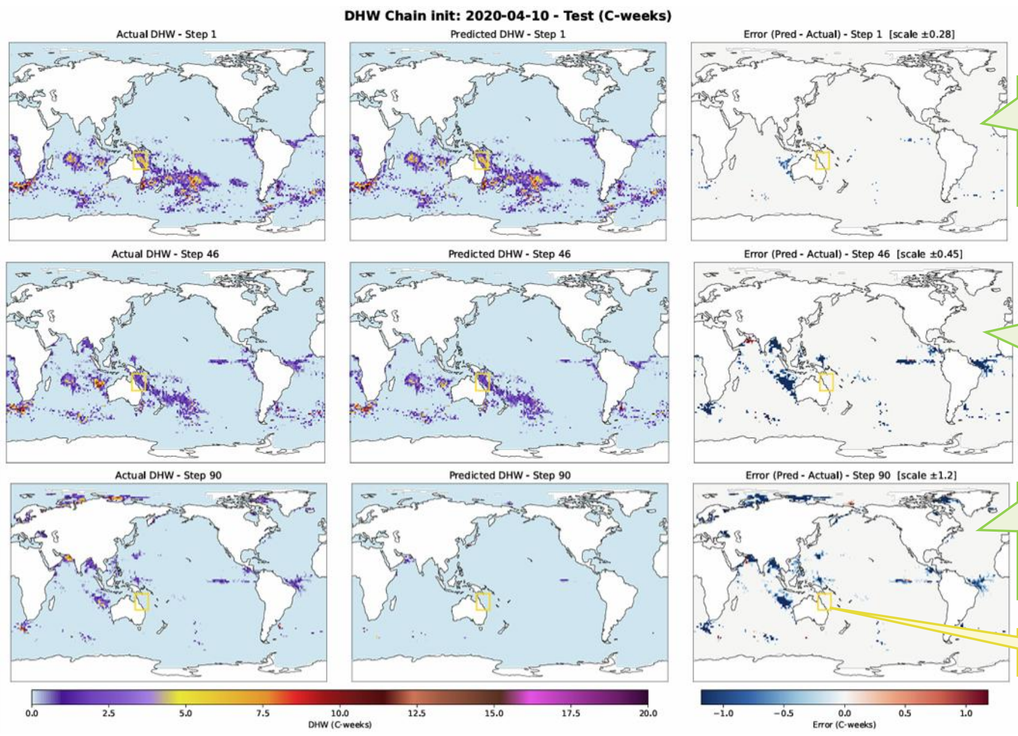
Day 1: Speckled noise along coasts and ocean fronts, where fine detail is hardest.

Day 46: Large-scale patterns still match closely; errors form broader warm and cool patches.

Day 90: No blow-up; global patterns hold as the forecast eases toward average conditions.

- How fast the model learns mattered most: tuning the learning schedule gave 6× better long-range pattern match.
- Training on longer forecast runs stops forecasts from collapsing beyond the training horizon.
- Finer maps made training unstable; no single setting fixed it.
- Best results came from a compact, stable configuration.

Degree Heating Weeks (DHW) Results



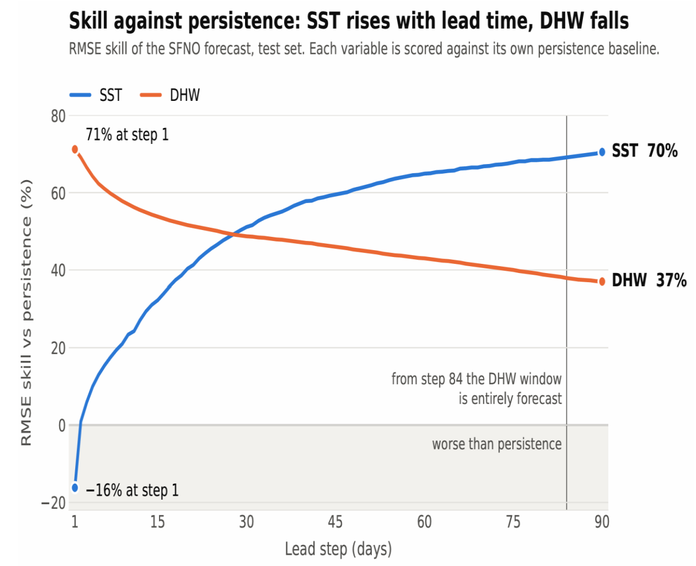
Day 1: the 12-week window is still almost all real data.

Day 46: Underestimating heat stress

Day 90: Underestimates heat stress

Great Barrier Reef

- Pattern match (ACC) stays above zero for all 90 days, and above 0.5 for ~69 days versus ~48 for persistence.
- By day 84, when the 12-week window is entirely forecast, DHW error is still ~40% lower than persistence.
- Best temperature model ≠ best heat-stress model: choosing models on temperature accuracy alone would have missed the best DHW forecaster.
- The models that best keep warm extremes, rather than smoothing them out, give the best heat-stress forecasts.



Percentage reduction in error compared with a persistence forecast. Above zero means better than persistence

Acknowledgements

This work was possible because of the work and support of my supervisor, John Taylor.

Australian National University

References

1. NOAA Coral Reef Watch, n.d., Daily Global 5km Product Methodology (v3.1), https://coralreefwatch.noaa.gov/product/5km/methodology.php (accessed 1 Oct 2026)
2. Skirving et al., 2019, Coral Reefs, https://doi.org/10.1007/s00338-019-01799-4
3. Grottoli et al., 2014, Global Change Biology, https://doi.org/10.1111/gcb.12658
4. Li et al., 2021, International Conference on Learning Representations (ICLR), https://doi.org/10.48550/arXiv.2010.08895